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WEB DEVELOPMENT5020

CHANGELOG

- I committed changes on my Part 1 POE (HTML) by adding comments in all my five pages, since my pages did not have any comments.
- According to the feedback I got from my lecturer about my pages not linking properly, I also got to fix the pages that were not linking.
- I also fixed my GitHub link.
- I also got to edit my sitemap.

CSS Changelog: Part 2

- The Paws & Hope Animal Rescue website was updated with an external CSS stylesheet to improve the appearance, consistency, usability and responsiveness of the website.

1. CSS Variables and Colour Scheme

Added reusable CSS variables using root.

Defined colours for:

- Main text (--ink)
- Secondary/muted text (--muted)
- Cream background (--cream)
- Orange (--orange)
- Dark orange (--orange-dark)
- Gold (--gold)
- Green (--green)
- White (--white)
- Borders (--line)
- Shadows (--shadow)
- Plum (--plum)

This makes the website colours consistent across all pages.

The variables also make future colour changes easier.

2. Global Box Sizing

- Added box-sizing: border-box to all elements.
- This makes the width and height of elements easier to control because padding and borders are included in the element's total size.

3. Smooth Scrolling

- Added scroll-behaviour: smooth.
- This allows smoother movement when navigating to sections within the page.

4. Body Styling

- Removed the default body margin.
- Added the main website text colour.
- Added a cream background colour.
- Set Arial, Helvetica and sans-serif as the main fonts.
- Set the font size to 16px.

- Added a line height of 1.65 to improve readability.

5. Image Styling

- Added responsive image sizing using:
- max width: 100%

height: auto

This prevents images from becoming wider than their containers while keeping their proportions.

6. Link Styling

- Added a consistent colour for website links.
- Links use the website's dark-orange colour.

7. Keyboard Focus Styling

Added: focus-visible styling to:

- Links
- Buttons
- Input fields
- Select menus
- Text areas

A visible gold outline appears when these elements receive keyboard focus.

This improves accessibility and makes it easier for keyboard users to see which element is selected.

8. Header Styling

- Made the header remain at the top of the screen while scrolling using position: sticky.
- Added a high z-index so the header stays above other content.
- Added a white background and subtle shadow.
- Styled the website branding area using Flexbox.
- Added spacing between the logo, website name and subtitle.

9. Logo and Brand Styling

- Set the logo to a consistent size.
- Prevented the logo from shrinking.
- Styled the main website heading.
- Styled the smaller description underneath the website name.

10. Navigation Bar

- Added a border above the navigation menu.
- Used Flexbox to arrange navigation links.
- Allowed navigation links to wrap when there is not enough horizontal space.
- Added spacing between navigation links.

- Removed the default bullet points from the navigation list.
- Styled navigation links with rounded corners.
- Added a hover effect that changes the link background to orange and the text to white.

11. Main Content

- Added a minimum height to the main content area.
- This helps keep the page layout balanced even when a page contains limited content.

12. Section Styling

- Limited the maximum width of the main sections to 1150px.
- Added automatic margins to centre the content.
- Added spacing between sections.
- Added consistent styling for section headings.
- Used responsive font sizing with clamp () for larger headings.

13. Hero Section

- Created a two-column layout using CSS Grid.
- One column contains the main introduction, and the other contains the hero image.
- Added spacing between the two columns.
- Made the hero heading responsive.
- Limited the width of the hero paragraph.
- Added rounded corners and a shadow to the hero image.

14. Page Images

- Added consistent styling for images used on other pages.
- Images are displayed at full available width within their containers.
- Added rounded corners.

Added object-fit: cover to keep images visually consistent.

Added maximum dimensions to prevent images from becoming excessively large.

15. Buttons

Created a consistent style for buttons and links that look like buttons.

Added:

- Rounded corners
- Orange background
- White text
- Padding
- Bold text
- Pointer cursor
- Added a hover effect that changes the button to dark orange.
- Added a small active effect when the button is pressed.

16. Card Layout

- Added a CSS Grid layout for cards.
- Large screens display three cards in a row.
- Added gaps between cards.

Styled card:

- White backgrounds
- Borders
- Rounded corners
- Shadows
- Added consistent sizing and cropping for card images.
- Added spacing and styling for card headings and paragraphs.

17. Form Styling

- Styled the enquiry/contact form as a separate content card.
- Limited the maximum form width to 760px.
- Added padding, borders, rounded corners and a shadow.
- Styled form labels.

Added consistent styling to:

- Text inputs
- Select fields
- Text areas
- Made text areas vertically resizable.
- Added spacing above the form button.

18. Form Message

- Styled #formMessage.
- Used the green colour and bold text to make feedback messages noticeable to the user.

19. Location Section

- Added a grid layout for location information.
- Used two columns on larger screens.

Styled each location section with:

- Padding
- Border
- Rounded corners
- White background

20. Footer

- Added a dark background to the footer.
- Changed footer text to white.
- Added padding around the footer.

- Centred the footer text.
- Removed unnecessary paragraph margins.
- Responsive Design

21. Tablet Layout – max-width: 800px

- Changed the hero section from two columns to one column.
- Reduced the hero image height.
- Changed the card layout from three columns to two columns.
- Centred the navigation links.

22. Mobile Layout – max-width: 560px

- Reduced header spacing.
- Reduced the logo size.
- Reduced the website title and subtitle sizes.
- Made navigation links smaller and more compact.
- Reduced hero section padding.
- Reduced the side padding of the main sections.
- Changed cards to a single-column layout.
- Changed location sections to a single-column layout.
- Reduced form padding to fit smaller screens.
- Accessibility and User Experience

23. Reduced Motion

- Added prefers-reduced-motion: reduce.
- Users who prefer less animation will have reduced transitions.
- Smooth scrolling is also disabled for these users.
- This improves accessibility for users who may be sensitive to excessive movement or animation.

CHANGELOG PART 3

1. Feedback from Part2

- I committed changes on my Part 2 POE (CSS) by fixing my header and footer to be the same throughout the pages.
- By following the feedback, I got from my lecturer I also worked on my navigation in all the pages.
- I changed the forms in all the pages since my pages had the same form.
- In addition, I also changed the colour of my font.

2. JavaScript Functionality

2.1 JavaScript File

- Created and linked "js/script.js" to the website pages. The script runs after the HTML has been parsed and uses the Document Object Model (DOM) to interact with page elements.

2.2 Image Lightbox

- Added an image lightbox that displays a larger version of selected gallery images.
- Added a close button and support for closing the lightbox by pressing Escape or clicking outside the image.
- Added keyboard activation and accessibility labels to gallery images.

2.3 Service Search and Sorting

- Added a search function that filters service cards according to the visitor's search text.
- Added alphabetical sorting in ascending and descending order.
- Added a result counter and a message when no services match the search.
- Used JavaScript DOM manipulation to update the displayed service cards.

2.4 FAQ Functionality

- Used HTML "details" and "summary" elements to allow visitors to expand and collapse frequently asked questions on the Services page.

2.5 Automatic Copyright Year

- Added JavaScript to display the current year in the footer, reducing the need to update it manually each year.

3. Form Validation and User Feedback

3.1 Enquiry Form

- Added JavaScript validation to the enquiry form.
- Checked required fields and email-format constraints.
- Added phone-number format validation.
- Included different guidance for adoption, fostering, volunteering and sponsorship enquiries.
- Added status messages to inform visitors whether their form passed validation or requires corrections.

3.2. Contact Form

- Added validation for required contact information and the general message.
- Added feedback messages to guide visitors when information is missing or invalid.
- Prepared a message containing the validated form information for the visitor's email application.

3.3 Form Limitations

- The forms use client-side validation and prepare an email draft addressed to "info@pawsandhope.co.za". The visitor must review and send the draft themselves. The website does not automatically send emails, store submissions on a server, process sponsorship payments, or confirm animal availability or adoption approval.
- A secure server-side endpoint or form-processing service would be required for automatic submission and storage.

The original combined JavaScript file was separated into six smaller files to make the code easier to understand, maintain and demonstrate in the assessment:

js/main.js: shared page functionality and automatic copyright year.

js/contact.js: contact form validation and email-draft preparation.

js/enquiry.js: enquiry form validation, phone-number checking and enquiry-type guidance.

js/lightbox.js: accessible image lightbox with mouse, keyboard and Escape-key controls.

js/map.js: Leaflet interactive map with an error fallback.

js/products.js: service search, filtering, result count and A–Z/Z–A sorting.

4. Interactive Map

- Added an interactive map to the Contact page using the Leaflet JavaScript library and OpenStreetMap tiles.
- Added a marker labelled for the Paws & Hope Animal Rescue assessment project.
- Included a fallback message and an OpenStreetMap link if the interactive map cannot load.
- Identified the marker as an approximate Polokwane city-centre location rather than a verified organisational address.

5.Search Engine Optimisation (SEO)

5.1 Page Metadata

- Added or reviewed page-specific HTML title tags and meta descriptions.
- Included relevant keywords in page metadata where appropriate.
- Reviewed heading structure to help visitors and search engines understand the content of each page.

5.2 Images and Internal Links

- Added or reviewed descriptive alternative text (alt) for images.
- Checked internal navigation links between the website pages.
- Used descriptive content to help explain the purpose of the website and its services.

5.3. Sitemap and Robots File

- Added a starter sitemap.xml file to list website pages for search-engine discovery.
- Added a robots.txt file to provide crawler instructions.
- Recorded the need to replace placeholder or relative sitemap information with the actual HTTPS website URL after deployment.

These files support SEO preparation but do not guarantee that search engines will index the website.

7. Documentation

- Updated the README documentation to explain the website structure and how to run the website locally.
- Documented the JavaScript features and form limitations.
- Added instructions for testing the website and identifying tasks that remain after deployment.
- Included references to documentation used for JavaScript, HTML form validation, SEO, maps and sitemap formatting.

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